

# Application of the Data Envelopment Analysis (DEA) to decision-making on labor and capital investment in the US hospitals

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## Abstract

This research applies an input- and output-oriented Data Envelopment Analysis (DEA) with a variable return to scale setting to decision making on investment in labor and capital in 63 US acute care hospitals. Depreciation and amortization, Salaries and wages as proxies for the capital and labor resources are the drivers. Patient accounts receivables and composite healthcare quality indicator PSI 90 risk-adjusted with the help of Patient Case Mix index are output variables and indicators of quantity and quality of the service provided. The analysis resulted in the following conclusions: 1. Potential reduction in labor and capital that is achievable without reduction of quantity and quality of the healthcare that varied from 0.09% to 43.31% per hospital. 2. Potential output increase that is achievable without increase in labor and capital investment that varied from 0.03% to 209.5% per hospital. 3. Risk-adjusted DEA is an effective method for evaluating hospital performance relative to peers and identifying opportunities for optimizing resource utilization.

**Key Words:** Data Envelopment Analysis, Healthcare, efficiency, optimization, capital, labor

## 1. Objective

The research is intended to provide a data driven and scientifically valid methodology for healthcare managers to answer the research questions:

- Does the hospital have any efficiency reserves in labor and capital resources to increase quantity and quality of healthcare without extra investment?
- Can the hospital maintain the same level of quantity and quality of the healthcare with less resources used?

## 2. Data

Table 2.1 below summarized information on publicly available data of the 63 US acute care hospitals for 2022 was used in the research.

## 3. Methodology

### 3.1. Risk adjustment of the healthcare service quality indicator PSI 90

PSI 90 is risk adjusted by applying ordinary least square regression with Case Mix Index as an only regressor to account for the patients' entry conditions:

- Expected  $PSI\ 90_i = \beta_0 + \beta_1 * CMI_i$  for the hospital  $i$
- Risk-adjusted  $PSI\ 90_i = -1 * (PSI\ 90_i - \text{Expected } PSI\ 90_i)$

where  $i = 1, \dots, 63$  hospital,

risk-adjusted  $PSI\ 90_i$  for the hospital is multiplied by -1 to imply that the better hospital serve its' patients, the lower negative deviations from the expected  $PSI\ 90_i$ .

### 3.2. Data Envelopment Analysis (DEA) with input and output orientation

The Data Envelopment Analysis (hereinafter, DEA) with input and output orientation are applied to focus on 2 different aspects of the hospital's efficiency of the healthcare. DEA is a non-parametric, linear programming technique used in operations research and economics for the estimation of production frontiers. It helps to empirically measure productive efficiency of decision making units or DMUs (Charnes, Cooper, and Rhodes (1978) that are hospitals in this research.

Table 2.1. Summary on the source data

| Data                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Role                               | Data source                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Depreciation and amortization                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Input, proxy for available capital | Maryland Health Services, Cost Review Submission, Hospital Financial Condition Report, Fiscal Year 2022, May 2023 |
| Salaries and wages                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Input, proxy for labor resources   | Maryland Health Services, Cost Review Submission, Hospital Financial Condition Report, Fiscal Year 2022, May 2023 |
| Patient accounts receivables                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Output                             | Maryland Health Services, Cost Review Submission, Hospital Financial Condition Report, Fiscal Year 2022, May 2023 |
| PSI 90 is a composite healthcare quality indicator that is aggregated from Medicare Fee-for-service including:<br>PSI 03 — Pressure Ulcer Rate;<br>PSI 06 — Iatrogenic Pneumothorax Rate<br>PSI 08 — In Hospital Fall-Associated Fracture Rate<br>PSI 09 — Postoperative Hemorrhage or Hematoma Rate<br>PSI 10 — Postoperative Acute Kidney Injury Requiring Dialysis Rate<br>PSI 11 — Postoperative Respiratory Failure Rate<br>PSI 12 — Perioperative Pulmonary Embolism or Deep Vein Thrombosis Rate<br>PSI 13 — Postoperative Sepsis Rate<br>PSI 14 — Postoperative Wound Dehiscence Rate<br>PSI 15 — Abdominopelvic Accidental Puncture/Laceration Rate | Output                             | Centers for Medicare and Medicaid Services                                                                        |
| Patient's Case Mix Index (CMI)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Instrumental variable              | Centers for Medicare and Medicaid Services                                                                        |

### 3.2.1. Input-Oriented DEA with Variable-Return to Scale (VRS)

This approach helps to answer the research question “Can the hospital maintain the same level of quantity and quality of the healthcare with less resources used?” (Charnes, A., Cooper, W., & Rhodes, E. (1978), Färe, R., Grosskopf, S., Lovell, C. (1994))

#### Definition

An input-oriented DEA model assesses how much each Decision Making Unit (DMU) (hospitals in this research) can reduce its inputs (costs/resources) while maintaining the same level of outputs (performance indicators).

#### Objective

Minimize inputs while maintaining output levels.

#### Mathematical Formulation

$$\begin{aligned}
 & \min_{\theta, \lambda} \quad \theta \\
 & \text{s.t.} \quad \sum_{j=1}^n \lambda_j x_{i,j} \leq \theta x_{i,o} \quad \forall i \\
 & \quad \sum_{j=1}^n \lambda_j y_{r,j} \geq y_{r,o} \quad \forall r \\
 & \quad \sum_{j=1}^n \lambda_j = 1 \quad (\text{VRS constraint}) \\
 & \quad \lambda_j \geq 0 \quad \forall j
 \end{aligned}$$

where

- $x_{i,j}$  = amount of input  $i$  used by DMU  $j$
- $y_{r,j}$  = amount of output  $r$  produced by DMU  $j$
- $\lambda_j$  = intensity weight assigned to peer DMU  $j$
- $\theta$  = scalar input efficiency score
- $\theta \in [0, 1]$ :
  - If  $\theta = 1$ : DMU is efficient (on the frontier)
  - If  $\theta < 1$ : DMU could **proportionally reduce all inputs** by  $(1 - \theta)$  and still produce the same outputs
- The VRS constraint  $\sum \lambda_j = 1$  allows for **variable returns to scale**

### 3.2.2. Output-oriented DEA model with variable returns to scale (VRS)

This approach helps to answer the question: “Does the hospital have any efficiency reserves to increase quantity and quality of healthcare without extra investment into the labor and capital?” (Charnes, A., Cooper, W. W., & Rhodes, E. (1978), Färe, R., Grosskopf, S., Lovell, C. (1994))

#### Definition

An output-oriented DEA model evaluates how much a Decision Making Unit (DMU) can proportionally increase its outputs given the same level of inputs.

#### Objective

Maximize outputs while keeping inputs constant.

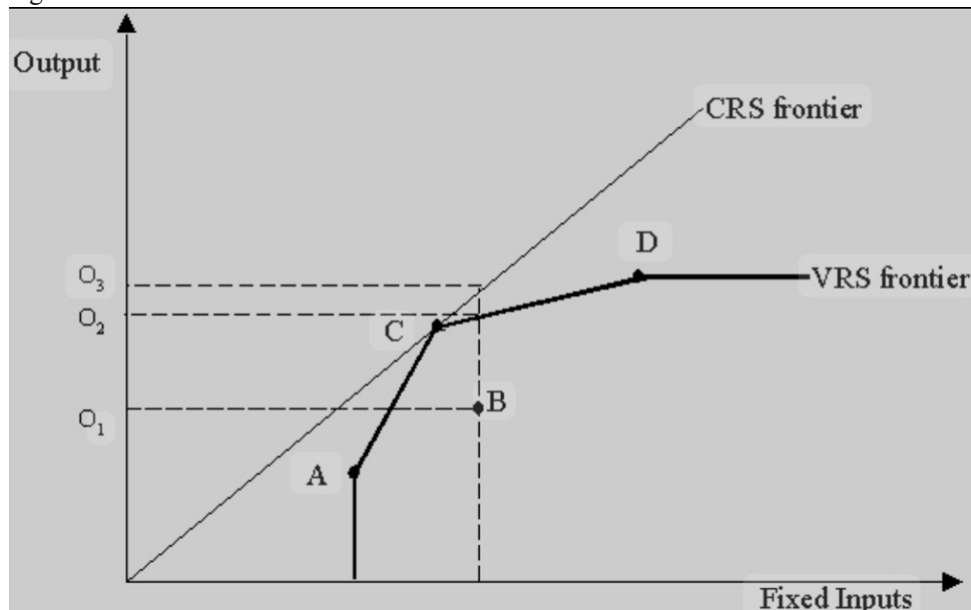
#### Mathematical Formulation

$$\begin{aligned}
& \max_{\phi, \lambda} \quad \phi \\
& \text{s.t.} \quad \sum_{j=1}^n \lambda_j x_{i,j} \leq x_{i,o} \quad \forall i \\
& \quad \quad \sum_{j=1}^n \lambda_j y_{r,j} \geq \phi \cdot y_{r,o} \quad \forall r \\
& \quad \quad \sum_{j=1}^n \lambda_j = 1 \quad (\text{VRS constraint}) \\
& \quad \quad \lambda_j \geq 0 \quad \forall j
\end{aligned}$$

- $x_{i,j}$ : input  $i$  used by DMU  $j$
- $y_{r,j}$ : output  $r$  produced by DMU  $j$
- $x_{i,o}, y_{r,o}$ : input and output of the **DMU under evaluation**
- $\lambda_j$ : weight assigned to DMU  $j$  (peer intensity)
- $\phi$ : output expansion factor (efficiency score)
- $\phi \geq 1$ : indicates by what factor the outputs of DMU  $o$  can be **proportionally expanded** without increasing inputs
- $\phi = 1$ : DMU lies on the **efficient frontier**
- $\phi > 1$ : DMU is **inefficient**, and could increase outputs by a factor of  $\phi$

The effect of the scale assumption on the measure of capacity utilization is demonstrated in Figure 3.1. Four data points (A, B, C, and D) are used to estimate the efficient frontier and the level of capacity utilization under both scale assumptions. Note that only fixed inputs are considered in Figure D.1. The frontier defines the full capacity output given the level of fixed inputs. With constant returns to scale, the frontier is defined by point C for all points along the frontier, with all other points falling below the frontier (hence indicating capacity underutilization). With variable returns to scale, the frontier is defined by points A, C and D, and only point B lies below the frontier i.e. exhibits capacity underutilization. The capacity output corresponding to variable returns to scale is lower than the capacity output corresponding to constant returns to scale (Food and Agriculture of the United Nations, Appendix D)

Figure 3.1. Visualization of the Contant Return and Variable Return to scale



This research applied both input- and output-oriented DEA with VRS to the following data.

Input variables (X):

- a) Depreciation and amortization
- b) Salaries and wages

Output variables (Y):

- a) Patient accounts receivable as an indicator of quantity of healthcare provided
- b) Risk adjusted PSI 90as an indicator of quality of healthcare provided

All data are obtained at the hospital level.

## 4. Results

Table 4.1 below presents results of the linear regression analysis by using statsmodels.api in Python. Marginal significance of the CMI effect at p-value of 0.129 is expected and good enough for the risk adjustment of the PSI 90. It can be explained by the data heterogeneity, aggregation at the hospital level, and limited sample size of 63 hospitals.

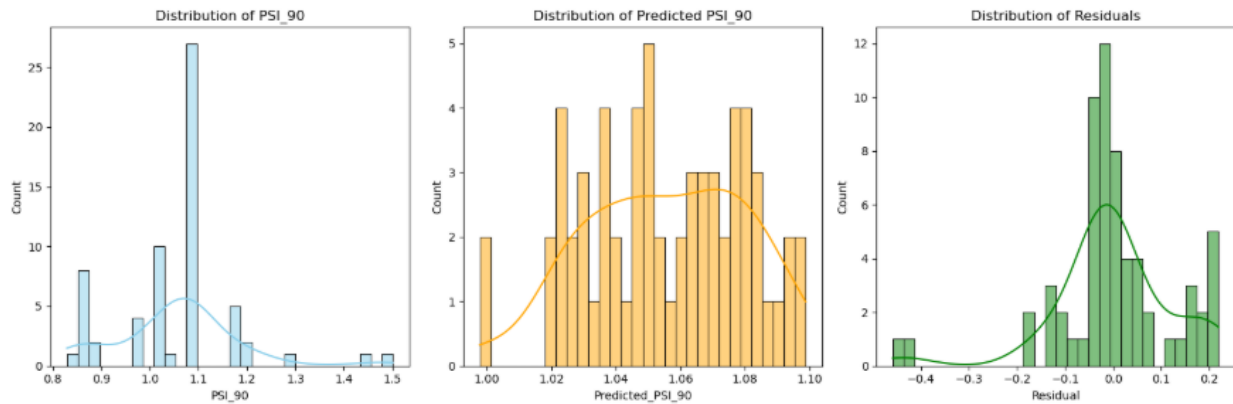
Tables 4.2 and 4.3 below presents results of the input- and output-oriented analysis implemented by using bootstrapped robust DEA with 1,000 iterations with dea.robust() in rDEA package in R.

Table 4.1. Regression modeling results for risk-adjustment of the PCI 90 by CMI

| OLS Regression Results |                  |                     |          |       |        |        |
|------------------------|------------------|---------------------|----------|-------|--------|--------|
| =====                  |                  |                     |          |       |        |        |
| Dep. Variable:         | PSI_90           | R-squared:          | 0.037    |       |        |        |
| Model:                 | OLS              | Adj. R-squared:     | 0.022    |       |        |        |
| Method:                | Least Squares    | F-statistic:        | 2.369    |       |        |        |
| Date:                  | Mon, 26 May 2025 | Prob (F-statistic): | 0.129    |       |        |        |
| Time:                  | 21:06:44         | Log-Likelihood:     | 42.107   |       |        |        |
| No. Observations:      | 63               | AIC:                | -80.21   |       |        |        |
| Df Residuals:          | 61               | BIC:                | -75.93   |       |        |        |
| Df Model:              | 1                |                     |          |       |        |        |
| Covariance Type:       | nonrobust        |                     |          |       |        |        |
| =====                  |                  |                     |          |       |        |        |
|                        | coef             | std err             | t        | P> t  | [0.025 | 0.975] |
| -----                  |                  |                     |          |       |        |        |
| const                  | 0.8827           | 0.113               | 7.789    | 0.000 | 0.656  | 1.109  |
| CMI                    | 0.1146           | 0.074               | 1.539    | 0.129 | -0.034 | 0.263  |
| =====                  |                  |                     |          |       |        |        |
| Omnibus:               | 18.718           | Durbin-Watson:      | 1.784    |       |        |        |
| Prob(Omnibus):         | 0.000            | Jarque-Bera (JB):   | 35.493   |       |        |        |
| Skew:                  | 0.944            | Prob(JB):           | 1.96e-08 |       |        |        |
| Kurtosis:              | 6.155            | Cond. No.           | 15.5     |       |        |        |
| =====                  |                  |                     |          |       |        |        |

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.



## 5. Conclusions

This research applied robust input and output-oriented Data Envelopment Analysis with variable return to scale to a dataset of 69 US hospitals including composite healthcare quality indicator PSI 90, Case Mix Index on patients' entry conditions, hospitals' expenses on salaries, wages and capital depreciation. The analysis resulted in the following conclusions:

1. Potential reduction in labor and capital that is achievable without reduction of quantity and quality of the healthcare that varied from 0.09% to 43.31% per hospital
2. Potential output increase that is achievable without increase in labor and capital investment that varied from 0.03% to 209.5% per hospital
3. Risk-adjusted DEA is an effective method for evaluating hospital performance relative to peers and identifying opportunities for optimizing resource utilization.

Table 4.2. Results of the input-oriented DEA with VRS

| Improve-<br>ment<br>potential<br>rank | Hospital<br>id | Bias-corrected<br>efficiency<br>score | 95% score<br>lower<br>bound | 95% score<br>upper<br>bound | Interpretation |                               |
|---------------------------------------|----------------|---------------------------------------|-----------------------------|-----------------------------|----------------|-------------------------------|
| 1                                     | 9              | 0.567                                 | 0.563                       | 0.577                       | 43.31%         | input reduction is achievable |
| 2                                     | 27             | 0.569                                 | 0.567                       | 0.573                       | 43.08%         | input reduction is achievable |
| 3                                     | 16             | 0.570                                 | 0.568                       | 0.573                       | 43.05%         | input reduction is achievable |
| 4                                     | 48             | 0.570                                 | 0.568                       | 0.573                       | 43.01%         | input reduction is achievable |
| 5                                     | 39             | 0.576                                 | 0.573                       | 0.580                       | 42.39%         | input reduction is achievable |
| 6                                     | 61             | 0.730                                 | 0.727                       | 0.737                       | 26.95%         | input reduction is achievable |
| 7                                     | 37             | 0.731                                 | 0.727                       | 0.738                       | 26.92%         | input reduction is achievable |
| 8                                     | 36             | 0.889                                 | 0.885                       | 0.896                       | 11.13%         | input reduction is achievable |
| 9                                     | 29             | 0.907                                 | 0.814                       | 1.566                       | 9.34%          | input reduction is achievable |
| 10                                    | 60             | 0.911                                 | 0.823                       | 1.577                       | 8.87%          | input reduction is achievable |
| 11                                    | 31             | 0.912                                 | 0.825                       | 1.573                       | 8.76%          | input reduction is achievable |
| 12                                    | 26             | 0.913                                 | 0.827                       | 1.576                       | 8.66%          | input reduction is achievable |
| 13                                    | 34             | 0.914                                 | 0.902                       | 0.964                       | 8.64%          | input reduction is achievable |
| 14                                    | 33             | 0.916                                 | 0.911                       | 0.927                       | 8.39%          | input reduction is achievable |
| 15                                    | 38             | 0.920                                 | 0.840                       | 1.589                       | 8.05%          | input reduction is achievable |
| 16                                    | 51             | 0.930                                 | 0.861                       | 1.612                       | 6.98%          | input reduction is achievable |
| 17                                    | 25             | 0.940                                 | 0.927                       | 0.983                       | 5.97%          | input reduction is achievable |
| 18                                    | 15             | 0.942                                 | 0.939                       | 0.948                       | 5.75%          | input reduction is achievable |
| 19                                    | 17             | 0.943                                 | 0.940                       | 0.948                       | 5.72%          | input reduction is achievable |
| 20                                    | 14             | 0.944                                 | 0.941                       | 0.949                       | 5.61%          | input reduction is achievable |
| 21                                    | 40             | 0.949                                 | 0.454                       | 1.647                       | 5.12%          | input reduction is achievable |
| 22                                    | 8              | 0.974                                 | 0.969                       | 0.984                       | 2.61%          | input reduction is achievable |
| 23                                    | 32             | 0.976                                 | 0.953                       | 1.077                       | 2.39%          | input reduction is achievable |
| 24                                    | 63             | 0.980                                 | 0.968                       | 1.030                       | 2.00%          | input reduction is achievable |
| 25                                    | 49             | 0.981                                 | 0.963                       | 1.032                       | 1.89%          | input reduction is achievable |
| 26                                    | 44             | 0.984                                 | 0.970                       | 1.026                       | 1.56%          | input reduction is achievable |
| 27                                    | 13             | 0.988                                 | 0.983                       | 0.996                       | 1.25%          | input reduction is achievable |
| 28                                    | 35             | 0.988                                 | 0.976                       | 1.002                       | 1.24%          | input reduction is achievable |
| 29                                    | 3              | 0.988                                 | 0.977                       | 1.039                       | 1.17%          | input reduction is achievable |
| 30                                    | 10             | 0.989                                 | 0.985                       | 0.996                       | 1.14%          | input reduction is achievable |
| 31                                    | 55             | 0.989                                 | 0.979                       | 1.029                       | 1.07%          | input reduction is achievable |
| 32                                    | 42             | 0.990                                 | 0.986                       | 1.000                       | 0.98%          | input reduction is achievable |
| 33                                    | 30             | 0.991                                 | 0.985                       | 1.006                       | 0.93%          | input reduction is achievable |
| 34                                    | 53             | 0.991                                 | 0.987                       | 0.998                       | 0.92%          | input reduction is achievable |
| 35                                    | 41             | 0.991                                 | 0.986                       | 1.005                       | 0.89%          | input reduction is achievable |
| 36                                    | 56             | 0.991                                 | 0.988                       | 0.999                       | 0.87%          | input reduction is achievable |
| 37                                    | 18             | 0.992                                 | 0.987                       | 1.003                       | 0.82%          | input reduction is achievable |
| 38                                    | 58             | 0.992                                 | 0.984                       | 1.020                       | 0.80%          | input reduction is achievable |
| 39                                    | 46             | 0.993                                 | 0.986                       | 1.006                       | 0.73%          | input reduction is achievable |
| 40                                    | 59             | 0.994                                 | 0.988                       | 1.008                       | 0.60%          | input reduction is achievable |

|    |    |       |       |       |       |                                   |
|----|----|-------|-------|-------|-------|-----------------------------------|
| 41 | 52 | 0.994 | 0.988 | 1.000 | 0.59% | input reduction is achievable     |
| 42 | 43 | 0.998 | 0.994 | 1.009 | 0.17% | input reduction is achievable     |
| 43 | 20 | 0.999 | 0.998 | 1.001 | 0.12% | input reduction is achievable     |
| 44 | 21 | 0.999 | 0.998 | 1.001 | 0.12% | input reduction is achievable     |
| 45 | 19 | 0.999 | 0.998 | 1.001 | 0.10% | input reduction is achievable     |
| 46 | 50 | 0.999 | 0.998 | 1.001 | 0.09% | input reduction is achievable     |
| 47 | 11 | 0.999 | 0.998 | 1.001 | 0.09% | input reduction is achievable     |
| 48 | 47 | 0.999 | 0.998 | 1.001 | 0.09% | input reduction is achievable     |
| 49 | 12 | 0.999 | 0.998 | 1.001 | 0.09% | input reduction is achievable     |
| 50 | 54 | 0.999 | 0.998 | 1.001 | 0.09% | input reduction is achievable     |
| 51 | 57 | 0.999 | 0.998 | 1.001 | 0.09% | input reduction is achievable     |
| 52 | 7  | 1.001 | 1.003 | 1.007 | 0.00% | input reduction is not achievable |
| 53 | 6  | 1.001 | 1.003 | 1.006 | 0.00% | input reduction is not achievable |
| 54 | 28 | 1.002 | 1.004 | 1.007 | 0.00% | input reduction is not achievable |
| 55 | 5  | 1.002 | 1.004 | 1.007 | 0.00% | input reduction is not achievable |
| 56 | 62 | 1.007 | 0.754 | 1.019 | 0.00% | input reduction is not achievable |
| 57 | 2  | 1.014 | 1.029 | 1.034 | 0.00% | input reduction is not achievable |
| 58 | 1  | 1.015 | 1.030 | 1.033 | 0.00% | input reduction is not achievable |
| 59 | 23 | 1.018 | 0.526 | 1.043 | 0.00% | input reduction is not achievable |
| 60 | 24 | 1.018 | 0.526 | 1.045 | 0.00% | input reduction is not achievable |
| 61 | 4  | 1.033 | 0.555 | 1.115 | 0.00% | input reduction is not achievable |
| 62 | 45 | 1.039 | 0.567 | 1.102 | 0.00% | input reduction is not achievable |
| 63 | 22 | 1.043 | 0.575 | 1.089 | 0.00% | input reduction is not achievable |

Table 4.3. Results of the output-oriented DEA with VRS

| Improve-<br>ment<br>potential<br>rank | Hospital<br>id | Bias-<br>corrected<br>efficiency<br>score | 95%<br>score<br>lower<br>bound | 95%<br>score<br>upper<br>bound | Interpretation |                                                              |
|---------------------------------------|----------------|-------------------------------------------|--------------------------------|--------------------------------|----------------|--------------------------------------------------------------|
| 1                                     | 46             | 3.19                                      | 2.98                           | 3.28                           | 209.5%         | output increase is achievable without increase in inputs     |
| 2                                     | 44             | 3.12                                      | 2.87                           | 3.37                           | 209.5%         | output increase is achievable without increase in inputs     |
| 3                                     | 55             | 3.18                                      | 2.96                           | 3.49                           | 209.5%         | output increase is achievable without increase in inputs     |
| 4                                     | 4              | 2.26                                      | 1.90                           | 2.66                           | 146.9%         | output increase is achievable without increase in inputs     |
| 5                                     | 5              | 2.37                                      | 2.19                           | 2.99                           | 146.9%         | output increase is achievable without increase in inputs     |
| 6                                     | 43             | 2.27                                      | 2.01                           | 2.68                           | 146.9%         | output increase is achievable without increase in inputs     |
| 7                                     | 2              | 2.34                                      | 2.16                           | 2.90                           | 146.9%         | output increase is achievable without increase in inputs     |
| 8                                     | 3              | 3.18                                      | 1.18                           | 8.29                           | 146.9%         | output increase is achievable without increase in inputs     |
| 9                                     | 7              | 2.38                                      | 2.23                           | 3.02                           | 146.9%         | output increase is achievable without increase in inputs     |
| 10                                    | 23             | 2.29                                      | 2.07                           | 2.76                           | 146.9%         | output increase is achievable without increase in inputs     |
| 11                                    | 24             | 2.27                                      | 2.03                           | 2.68                           | 146.9%         | output increase is achievable without increase in inputs     |
| 12                                    | 45             | 2.26                                      | 1.97                           | 2.66                           | 146.9%         | output increase is achievable without increase in inputs     |
| 13                                    | 52             | 2.29                                      | 2.07                           | 2.74                           | 146.9%         | output increase is achievable without increase in inputs     |
| 14                                    | 62             | 2.31                                      | 2.10                           | 2.79                           | 146.9%         | output increase is achievable without increase in inputs     |
| 15                                    | 16             | 1.32                                      | 1.04                           | 1.50                           | 71.4%          | output increase is not achievable without increase in inputs |
| 16                                    | 27             | 1.34                                      | 1.06                           | 1.60                           | 71.4%          | output increase is not achievable without increase in inputs |
| 17                                    | 48             | 1.31                                      | 1.04                           | 1.48                           | 71.4%          | output increase is not achievable without increase in inputs |
| 18                                    | 9              | 1.37                                      | 0.80                           | 1.66                           | 67.8%          | output increase is not achievable without increase in inputs |
| 19                                    | 39             | 1.25                                      | 0.98                           | 1.43                           | 66.0%          | output increase is not achievable without increase in inputs |
| 20                                    | 37             | 0.97                                      | 0.75                           | 1.11                           | 34.4%          | output increase is not achievable without increase in inputs |
| 21                                    | 61             | 0.97                                      | 0.75                           | 1.11                           | 34.4%          | output increase is not achievable without increase in inputs |
| 22                                    | 36             | 0.87                                      | 0.70                           | 0.99                           | 11.5%          | output increase is not achievable without increase in inputs |
| 23                                    | 34             | 1.14                                      | 0.73                           | 1.29                           | 8.0%           | output increase is not achievable without increase in inputs |
| 24                                    | 25             | 1.20                                      | 0.74                           | 1.35                           | 7.4%           | output increase is not achievable without increase in inputs |
| 25                                    | 33             | 0.98                                      | 0.76                           | 1.56                           | 7.1%           | output increase is not achievable without increase in inputs |
| 26                                    | 15             | 0.98                                      | 0.88                           | 1.10                           | 4.8%           | output increase is not achievable without increase in inputs |
| 27                                    | 17             | 0.91                                      | 0.78                           | 1.62                           | 4.7%           | output increase is not achievable without increase in inputs |
| 28                                    | 18             | 0.79                                      | 0.61                           | 1.06                           | 4.7%           | output increase is not achievable without increase in inputs |
| 29                                    | 14             | 0.98                                      | 0.88                           | 1.10                           | 4.6%           | output increase is not achievable without increase in inputs |
| 30                                    | 8              | 0.93                                      | 0.72                           | 1.98                           | 1.7%           | output increase is not achievable without increase in inputs |
| 31                                    | 40             | 1.04                                      | 0.96                           | 1.10                           | 1.5%           | output increase is not achievable without increase in inputs |
| 32                                    | 10             | 0.71                                      | 0.55                           | 0.81                           | 0.7%           | output increase is not achievable without increase in inputs |
| 33                                    | 13             | 0.91                                      | 0.80                           | 1.02                           | 0.7%           | output increase is not achievable without increase in inputs |
| 34                                    | 63             | 0.94                                      | 0.70                           | 2.20                           | 0.6%           | output increase is not achievable without increase in inputs |
| 35                                    | 53             | 0.71                                      | 0.54                           | 0.81                           | 0.5%           | output increase is not achievable without increase in inputs |

|    |    |      |      |       |      |                                                              |
|----|----|------|------|-------|------|--------------------------------------------------------------|
| 36 | 54 | 0.66 | 0.49 | 0.69  | 0.5% | output increase is not achievable without increase in inputs |
| 37 | 42 | 0.90 | 0.79 | 1.11  | 0.5% | output increase is not achievable without increase in inputs |
| 38 | 56 | 0.68 | 0.52 | 0.80  | 0.4% | output increase is not achievable without increase in inputs |
| 39 | 57 | 0.65 | 0.48 | 0.70  | 0.4% | output increase is not achievable without increase in inputs |
| 40 | 41 | 0.89 | 0.78 | 1.10  | 0.3% | output increase is not achievable without increase in inputs |
| 41 | 30 | 0.88 | 0.75 | 1.06  | 0.3% | output increase is not achievable without increase in inputs |
| 42 | 1  | 1.00 | N/a  | N/a   | 0.0% | output increase is not achievable without increase in inputs |
| 43 | 6  | 0.62 | 0.61 | 0.66  | 0.0% | output increase is not achievable without increase in inputs |
| 44 | 11 | 0.66 | 0.49 | 0.69  | 0.0% | output increase is not achievable without increase in inputs |
| 45 | 12 | 0.64 | 0.48 | 0.66  | 0.0% | output increase is not achievable without increase in inputs |
| 46 | 19 | 0.67 | 0.50 | 0.75  | 0.0% | output increase is not achievable without increase in inputs |
| 47 | 20 | 0.65 | 0.49 | 0.70  | 0.0% | output increase is not achievable without increase in inputs |
| 48 | 21 | 0.64 | 0.48 | 0.66  | 0.0% | output increase is not achievable without increase in inputs |
| 49 | 22 | 0.64 | 0.49 | 0.65  | 0.0% | output increase is not achievable without increase in inputs |
| 50 | 28 | 0.63 | 0.56 | 0.67  | 0.0% | output increase is not achievable without increase in inputs |
| 51 | 29 | 1.86 | 0.46 | 13.10 | 0.0% | output increase is not achievable without increase in inputs |
| 52 | 31 | 1.17 | 0.68 | 1.39  | 0.0% | output increase is not achievable without increase in inputs |
| 53 | 32 | 0.84 | 0.51 | 1.22  | 0.0% | output increase is not achievable without increase in inputs |
| 54 | 35 | 0.75 | 0.58 | 0.85  | 0.0% | output increase is not achievable without increase in inputs |
| 55 | 38 | 0.86 | 0.54 | 1.53  | 0.0% | output increase is not achievable without increase in inputs |
| 56 | 47 | 0.62 | 0.61 | 0.64  | 0.0% | output increase is not achievable without increase in inputs |
| 57 | 49 | 0.87 | 0.73 | 1.05  | 0.0% | output increase is not achievable without increase in inputs |
| 58 | 50 | 0.69 | 0.51 | 0.77  | 0.0% | output increase is not achievable without increase in inputs |
| 59 | 60 | 1.08 | 0.83 | 1.17  | 0.0% | output increase is not achievable without increase in inputs |
| 60 | 26 | 1.14 | 0.69 | 1.31  | 0.0% | output increase is not achievable without increase in inputs |
| 61 | 59 | 1.06 | 0.67 | 1.12  | 0.0% | output increase is not achievable without increase in inputs |
| 62 | 58 | 1.07 | 0.68 | 1.15  | 0.0% | output increase is not achievable without increase in inputs |
| 63 | 51 | 1.73 | 0.55 | 6.15  | 0.0% | output increase is not achievable without increase in inputs |

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